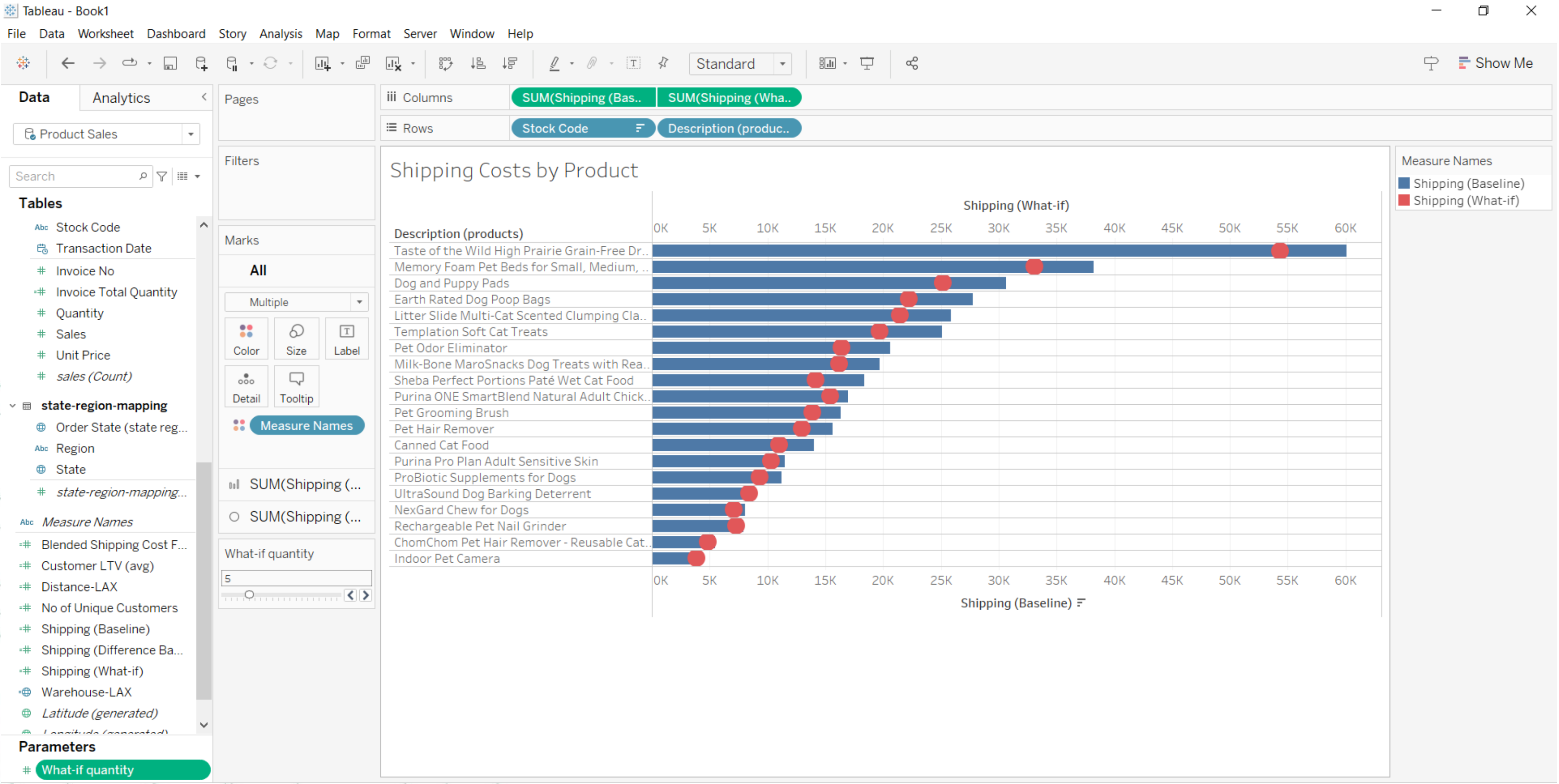
Step 35– Shipping Cost Analysis

- Using the fields created in the prior step, let us compare shipping costs (baseline) with a hypothetical what-if scenario.
- Change parameter values and see how it impacts shipping costs.
- On a new sheet, **"Shipping Costs by Product"**,
- create a **horizontal bar chart** displaying **Stock Code**, **Product Description**, and **Shipping (Baseline)** metrics.
- Add the Shipping (What-if) metric on a dual axis.
- Synchronize Axis and change the mark type of Shipping (What-if) to a Circle.
- Display the What-if Quantity parameter and observe how it impacts the measure values.
- **What will be the shipping cost savings (the difference between baseline and what-if shipping amount) for Dog and Puppy Pads when the What-if Quantity is set at five? If 5,344 proceed**
- That's correct. As per the **Blended Shipping Cost Factor**, when the shipped quantity is **five**, each quantity above one only costs **50% of the unit shipping cost**.
- The **Shipping (Baseline)** amount is **$30,492**.
- The **Shipping (What-if)** amount is **$25,148**.

Step 36– Shipping Cost Analysis

Adjust your visual to be the same.



Step 37– Shipping Cost Analysis

- The management wants to see a **dashboard** displaying shipping metrics and an ability to analyze what-ifs.

Here are the business requirements:

• **Prominently display key KPIs**. They are sales and cumulative shipping costs (baseline, what-if, and difference).

•**Change shipped quantity using a parameter** and an action filter on the product.

•**Breakdown of baseline and what-if shipping costs by product**.

- Display **key KPIs in a large font**. They are also called **Big Angry Numbers (BANs)**, and help answer important questions right away.

Your dashboard will need to display **four KPIs in large numbers**.

- Create **four new sheets** and visualize **sales** and the three main **shipping metrics** (baseline, what-if, and difference baseline).

• Name the sheets: **"BAN-Sales"**, **"BAN-Shipping Baseline"**, **“BAN-Shipping What-IF” and “BAN-Shipping Difference”**.

Step 38– Shipping Cost Analysis

Create a New Sheet - "Shipping What-if"

•Display the **running total** of the three main shipping metrics in an **area chart with transaction month**.

•Apply **color** to differentiate metrics.

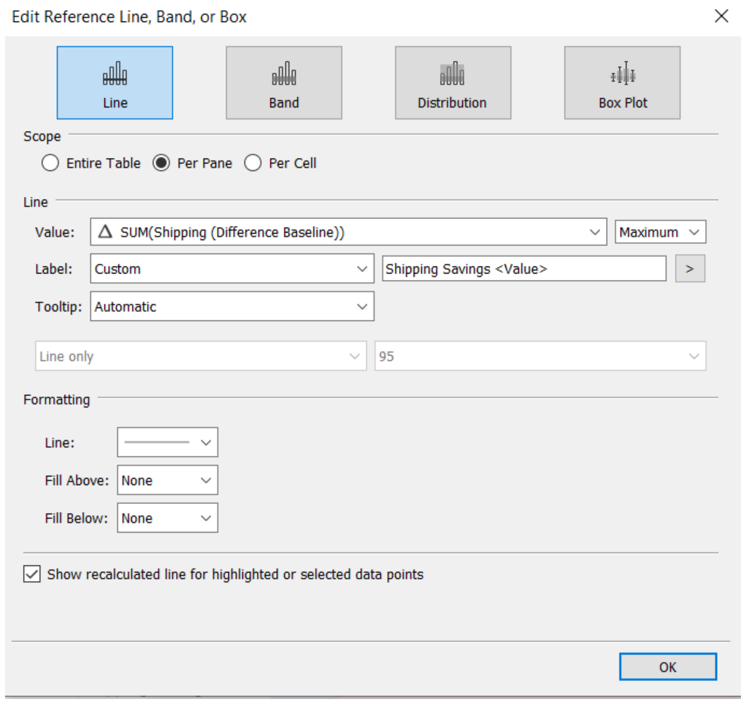
Add a Reference Line

•Insert a **reference line** on the **area chart**.

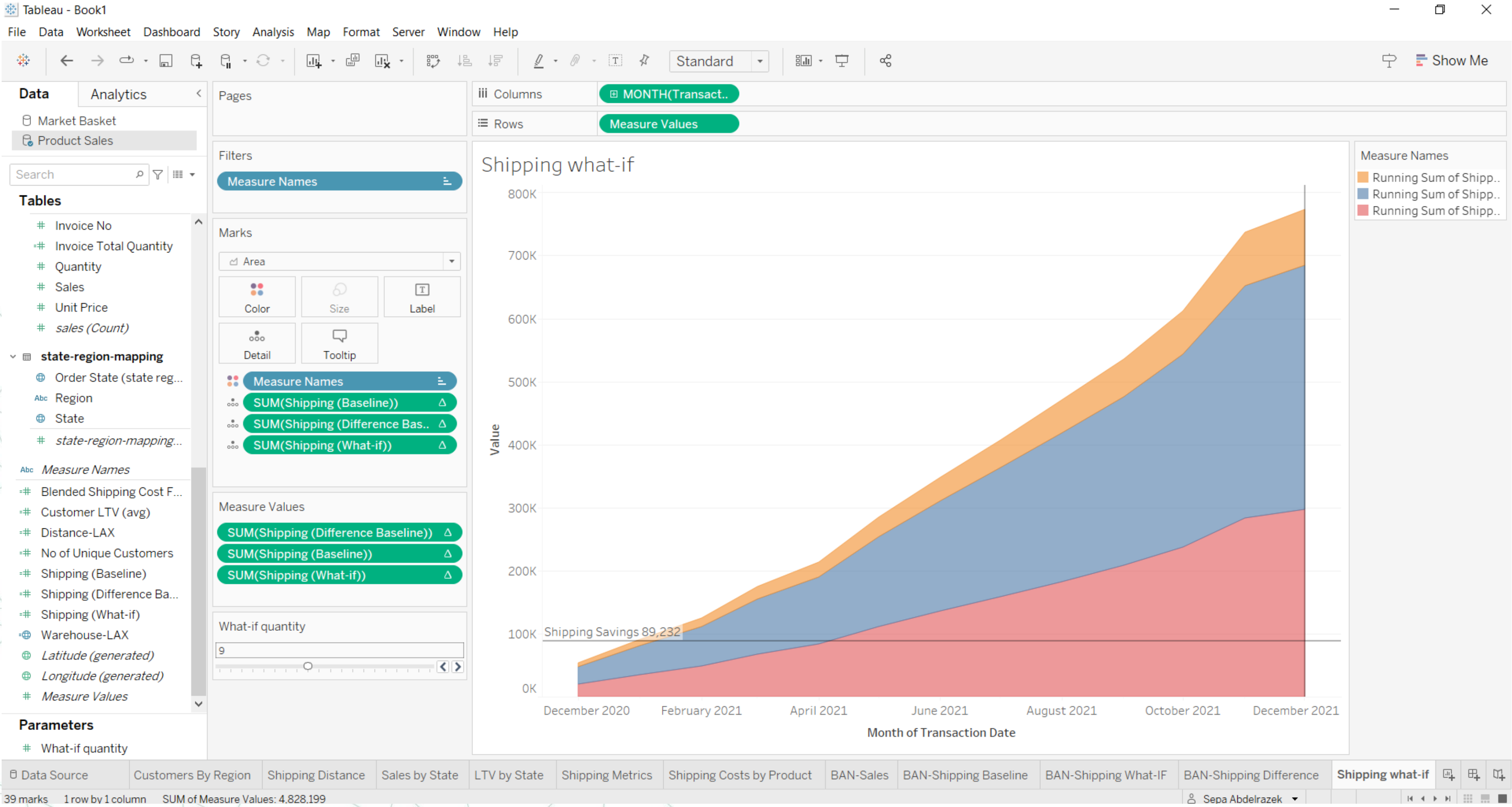
•Add a **custom label: "Shipping savings"**.

•Include the **running total of Shipping (Difference Baseline)** in the reference line.

•Please navigate to next slide to see how it should be



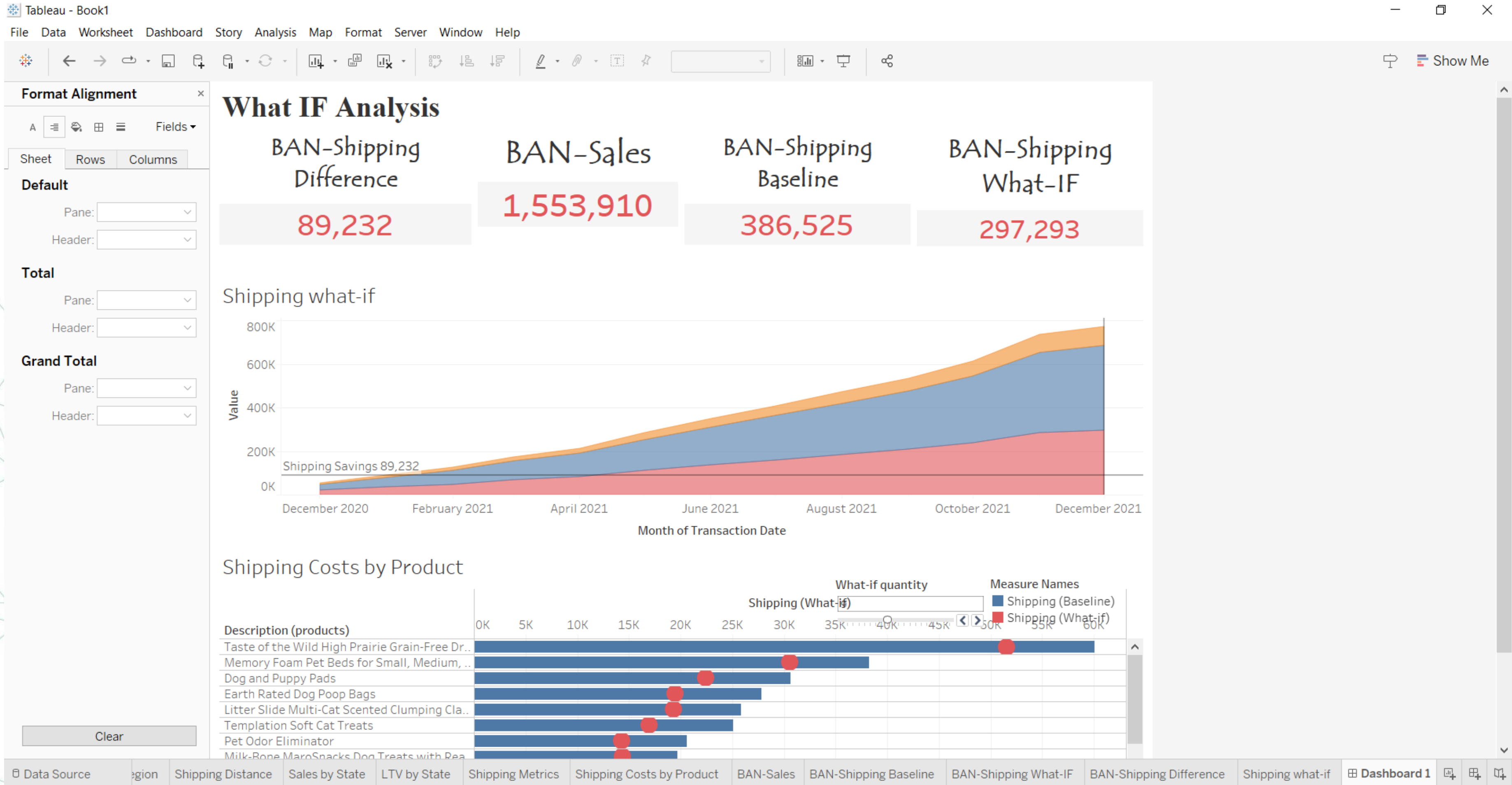
Step 39– Shipping Cost Analysis



Step 40– Shipping Cost Analysis

- **Create the What-If Analysis Dashboard**
- **Visualize the What-If Dashboard ("What-If Analysis")** using:
    - **Four big KPIs (BANs)**
    - **Area chart (Shipping What-if)**
    - **Shipping Costs by Product** sheet.
- Set the dashboard size to **at least 1200×1000** or as per preference.
- **Ensure the What-If parameter is displayed** and enable **Total Shipping Costs by Product** as an **action filter**.

Step 40– Shipping Cost Analysis



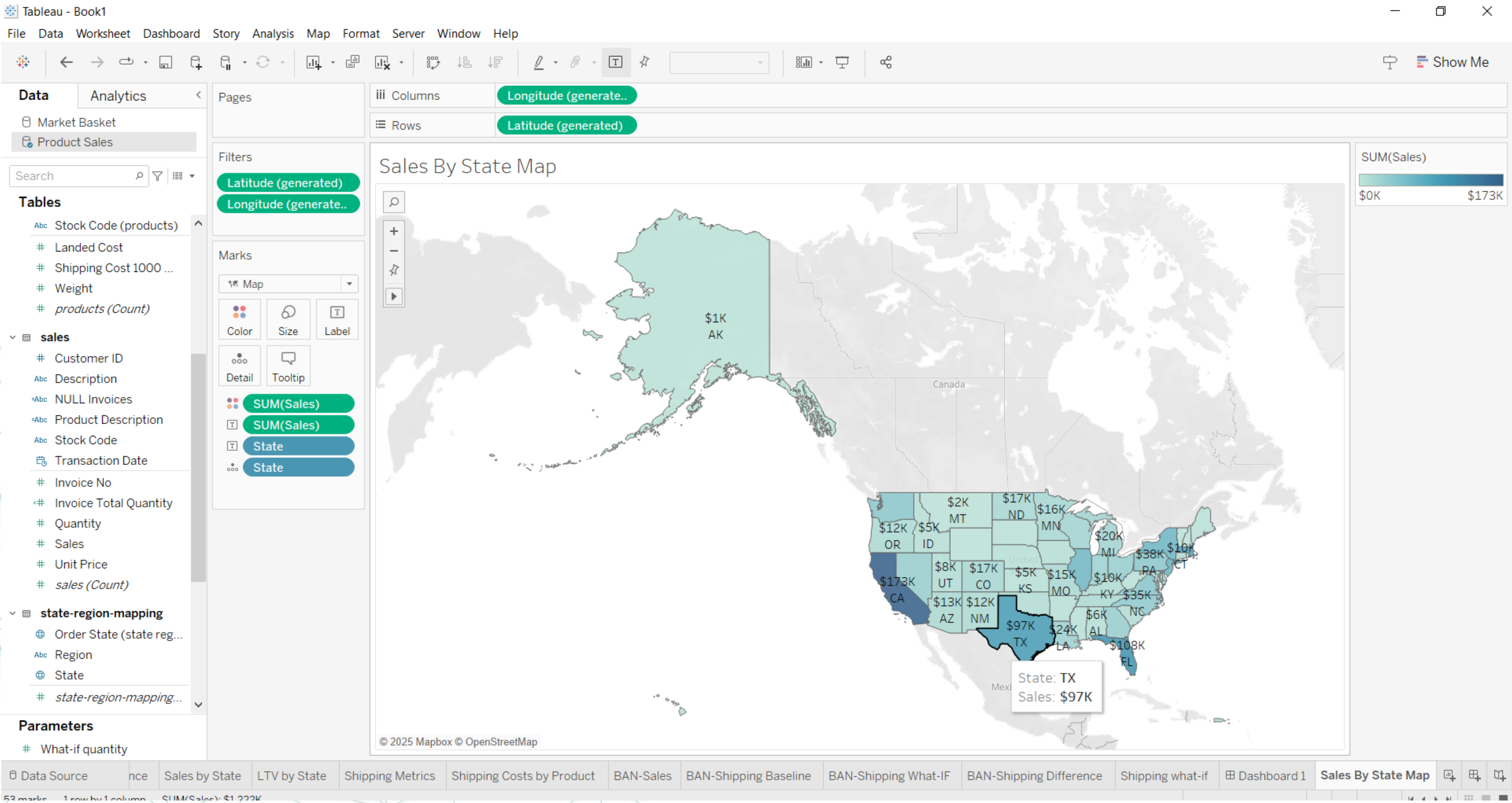
Step 41– Data Visualization & Storytelling

- Your data story will consist of three dashboards.
- You will reuse many of the charts built in the previously.
- In the **executive summary dashboard**, you will provide key KPIs such as sales and profit. This dashboard will provide a high-level pulse of the business.
- **The growth opportunities dashboard** will provide specific recommendations on how we can grow sales using cross-sell promotions.
- **The shipping costs dashboard** will suggest multiple strategies to reduce shipping costs.

Step 42– Data Visualization & Storytelling

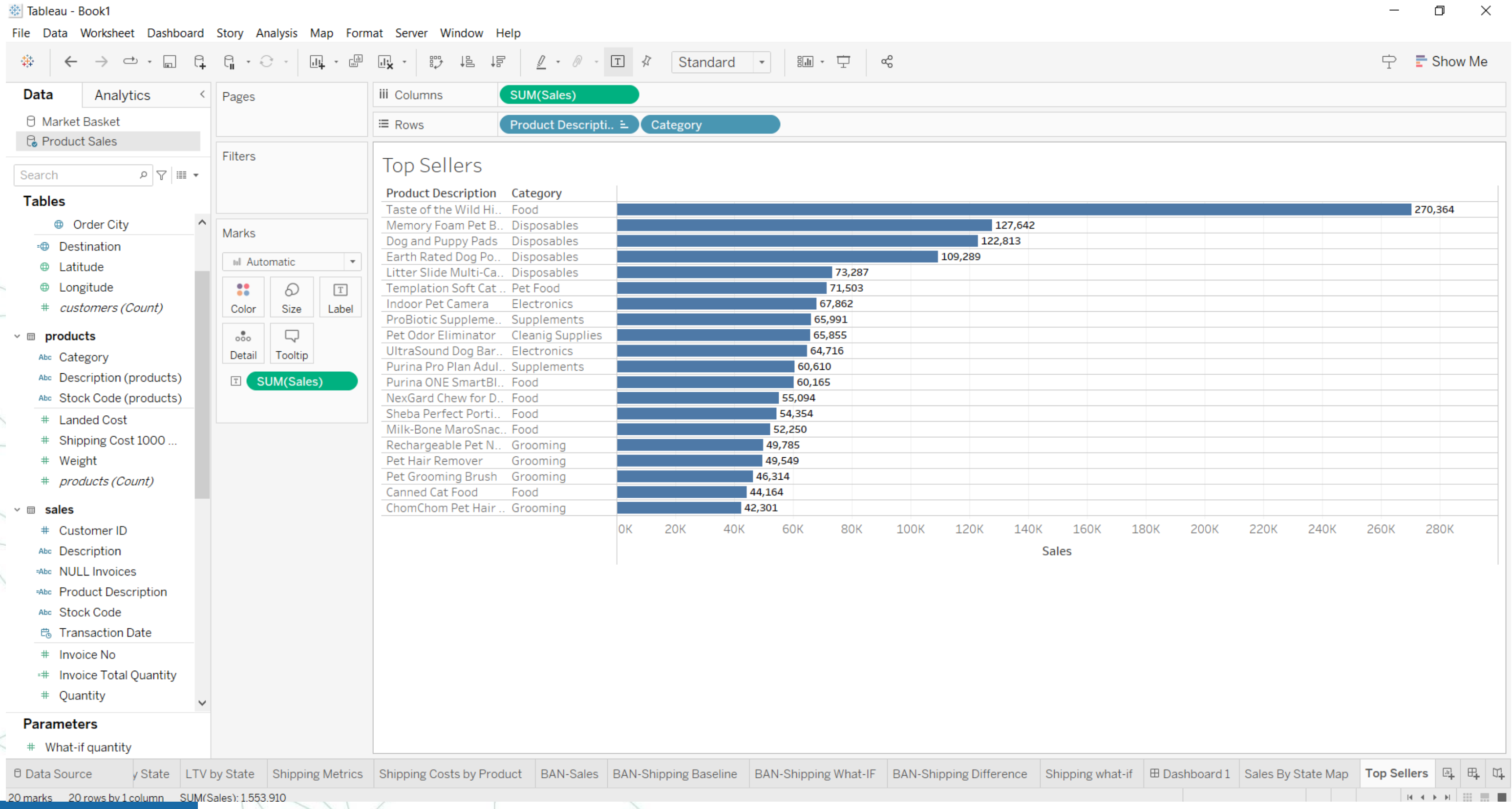
- Revenue by location and top sellers
- Let us build core KPIs which executives frequently ask to see.
- For example, what are my top-selling products, and who are my top customers? To answer this, using the **Product Sales** data source, you will build a map of sales by state and a bar chart for top-selling products.
- **Create a new sheet, "Sales by State Map".**
- Display a map with states colored by the amount of sales and add the Sales amount as a label.
- Display the sales amount in thousands (integer), with the US dollar currency symbol.

Step 42– Data Visualization & Storytelling



Step 43– Data Visualization & Storytelling

- **Create a new sheet, "Top sellers".**
- Create a horizontal bar chart of the amount of sales by Product Description and Category.
- Sort descending on the Sales amount and display the sales amount as a label.



Step 44– Data Visualization & Storytelling

- **KPIs in Big Angry Numbers (BANs)**
- You will be creating **BANs (Big Angry Numbers)** similar to those you created previously.
- You will build three different BANs: **Profit, Profit Percentage, and Shipping Baseline**.
- **Create two new calculated fields.**
- The First calculated field COGS (cost of goods sold) as the multiplication of Landed Cost and Quantity.
- **Create Profit (Baseline) as sales subtracted by COGS and Shipping (Baseline).**
- **Set the default** number format for both fields to a US dollar integer value.
- **Create a third field, Profit %, and define it as: the sum of Profit (Baseline) divided by the sum of Sales.**
- Set the number format as a percentage with two decimal points.
- **Create a new sheet, "BAN-Profit", and display Profit (Baseline) in a large font.**
- Format the sheet to remove all lines and borders.
- **Create two new sheets, "BAN-Profit Percentage" and "BAN-Shipping Baseline", by duplicating BAN-Profit and replacing: Profit (Baseline) with Profit %. Profit (Baseline) with Shipping (Baseline) respectively.**
- **What is the profit amount on the "BAN-Profit Percentage" sheet? If 27% you are on the right track**

Step 45– Data Visualization & Storytelling

- **Build an Executive Summary.**

We now have all the pieces to create an executive summary. An executive summary provides a quick pulse of the business operations and displays metrics such as sales, profit, and expenses. In addition, it provides an ability to drill down or filter on dimensions like product and customer location.

- **Create a new dashboard**, **Executive Summary**, with a custom size of at least 1200px width and 1000px height.
- Add the **BAN-Sales, BAN-Profit, BAN-Profit Percentage, and BAN-Shipping Baseline**
- **Replace titles, align text, and ensure all four sheets are evenly aligned.**
- **Add Top sellers, Sales by State Map, and LTV by State to the dashboard.**
- Adjust titles, alignment, and formatting.
- Apply a filter on Product Description to all sheets on the dashboard and use the map as an action filter.

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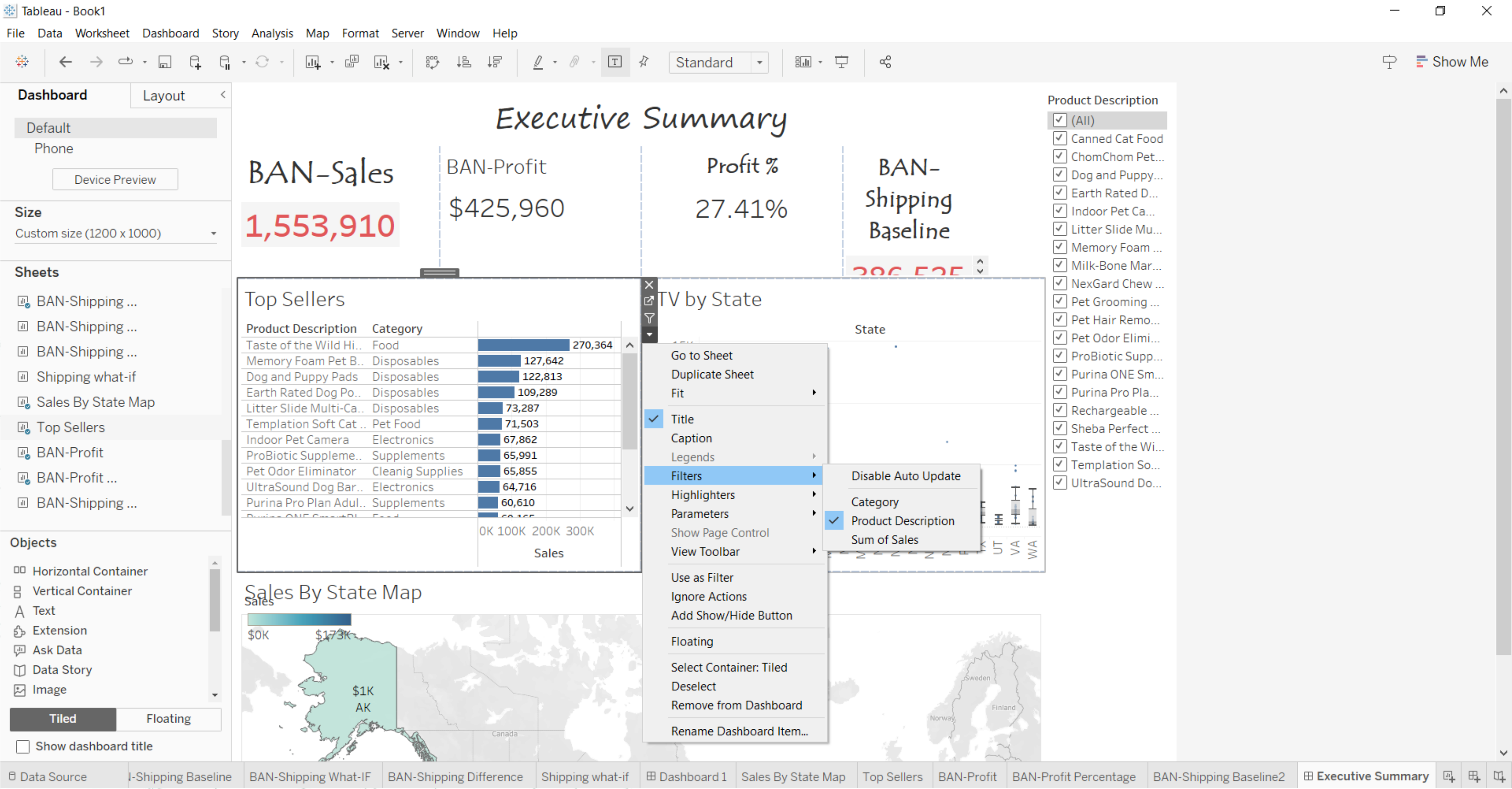
Step 45– Data Visualization & Storytelling

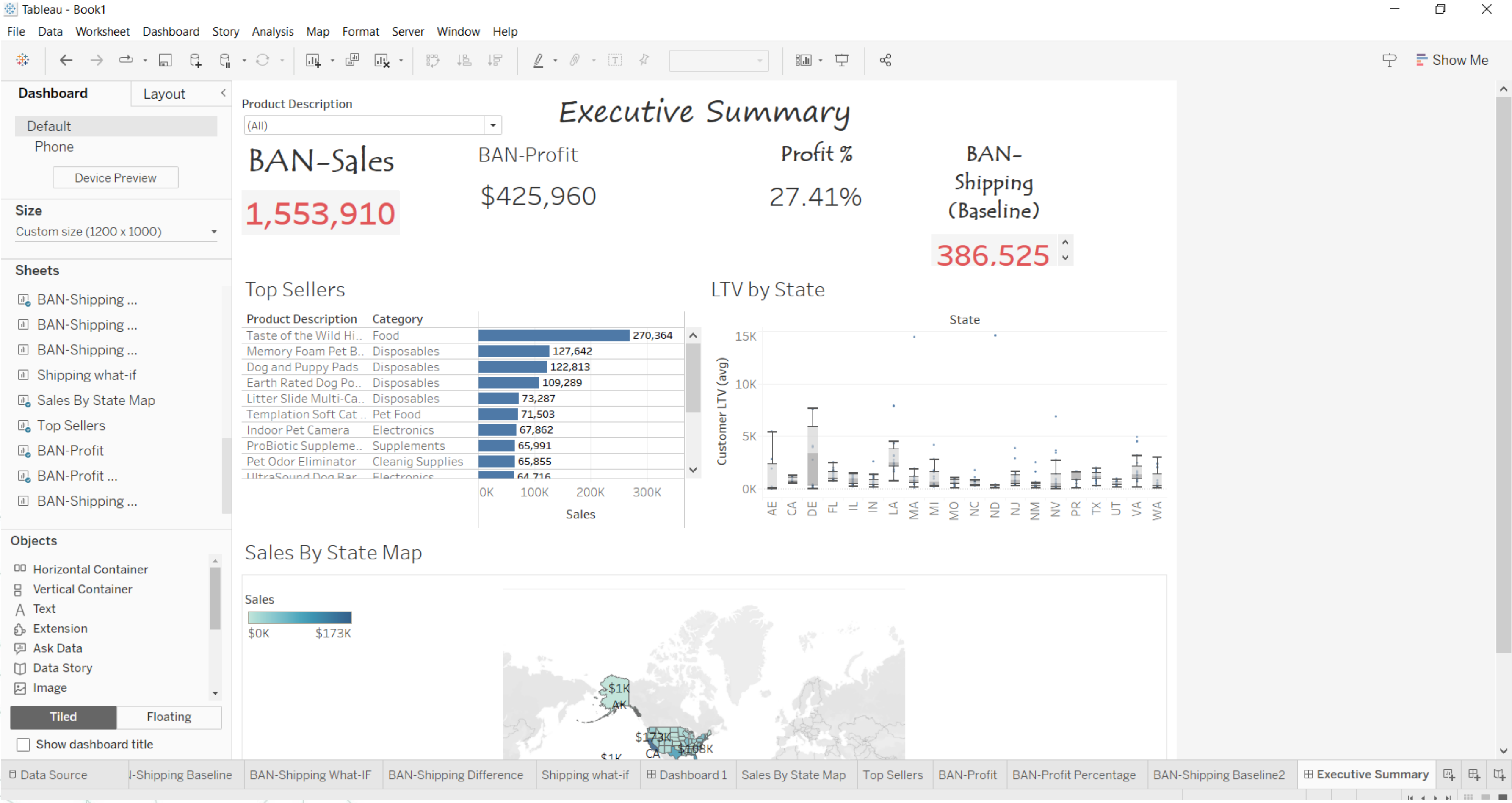


Step 46– Data Visualization & Storytelling

Align Your Dashboard

What is the profit % margin for Taste of the Wild High Prairie dog food in the state of California (code: CA)?

IF 26.26% proceed



Step 46– Data Visualization & Storytelling

Align Your Dashboard

What is the profit % margin for Taste of the Wild High Prairie dog food in the state of California (code: CA)?

IF 26.26% proceed